plot_trait_group_distribution.R

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Contents

`plot_trait_group_distribution` *Plot Raw Trait Value Distribution for One Taxon-Domain Group*

Description

Builds a ‘ggplot2‘ strip + boxplot showing every raw trait value for a single ‘taxon_name x trait_domain_name‘ group, with colour-coded Tukey fence flags and horizontal fence lines for both the standard (1.5x IQR) and extreme (3x IQR) thresholds.

Usage

```
plot_trait_group_distribution(
  data_group_raw,
  data_group_summary,
  sel_taxon,
  sel_domain,
  graphical_options,
  verbose = TRUE
)
```

Arguments

`data_group_raw` A tibble of raw trait observations for one taxon x domain group. Must contain columns ‘trait_value‘ (numeric), ‘trait_name‘ (character), and ‘trait_domain_name‘ (character).

`data_group_summary` A single-row tibble of per-group QC statistics, as produced by ‘generate_trait_qc_report()‘. Must contain numeric columns ‘mean‘, ‘median‘, ‘IQR‘, and integer columns ‘n_suspected_outliers_taxon‘ (integer) and ‘outlier_fraction‘ (numeric).

`sel_taxon` Character scalar. Name of the taxon being inspected. Used in the plot title.

sel_domain
: Character scalar. Name of the trait domain being inspected. Used in the plot title.

graphical_options
: Named list with elements ‘width‘, ‘height‘, ‘units‘, ‘dpi‘, and ‘bg‘, as returned by ‘load_active_config_value("graphical")‘. Passed to ‘ggview::canvas()‘.

verbose
: Logical. If ‘TRUE‘ (default), the computed Tukey fence boundaries are printed to the console via ‘cli::cli_inform()‘.

Details

The function computes Q1, Q3, and IQR from ‘data_group_raw‘ and derives four fence values: - inner lower / upper: Q1 - 1.5 * IQR and Q3 + 1.5 * IQR - outer lower / upper: Q1 - 3 * IQR and Q3 + 3 * IQR

Each observation is classified as ‘"within fence"‘, ‘"mild outlier (1.5x IQR)"‘, or ‘"extreme outlier (3x IQR)"‘. When ‘data_group_raw‘ contains more than one distinct ‘trait_name‘, observations are faceted by ‘trait_name‘; otherwise the x-axis shows ‘trait_domain_name‘.

Plot layer order follows project conventions: ‘ggplot()‘ -> facets -> scales -> labels -> theme -> ‘ggview::canvas()‘ -> geoms.

Value

A ‘ggplot2‘ object. The plot is not printed; call ‘print()‘ on the return value to display it.

See Also

[generate_trait_qc_report()], [load_trait_corrections()], [validate_trait_corrections()], [correct_trait_records()]

Examples

```
## Not run:
data_raw <-
  data_traits_raw |>
  dplyr::filter(
    taxon_name == "Anacyclus clavatus",
    trait_domain_name == "Leaf Area"
  ) |>
  dplyr::arrange(trait_value)

data_summary <-
  data_qc_report |>
  dplyr::filter(
    taxon_name == "Anacyclus clavatus",
    trait_domain_name == "Leaf Area"
  )

graphical_options <-
  load_active_config_value("graphical")

p <-
  plot_trait_group_distribution(
    data_group_raw = data_raw,
    data_group_summary = data_summary,
    sel_taxon = "Anacyclus clavatus",
    sel_domain = "Leaf Area",
    graphical_options = graphical_options
```

```
  )

base::print(p)

## End(Not run)
```

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